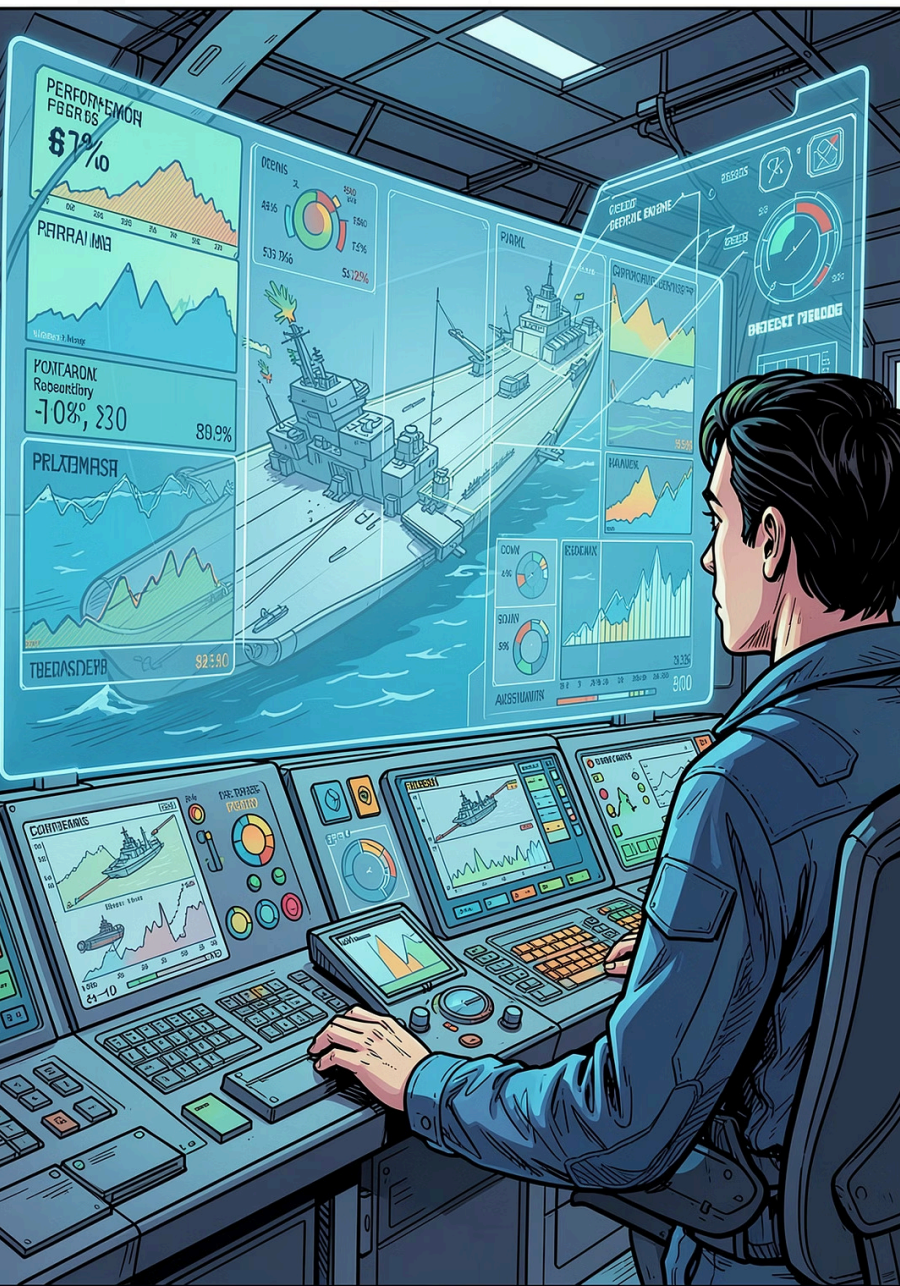




Impact & Success Metrics

Measuring the performance, safety, and business value of EMMA's local agent fleet.

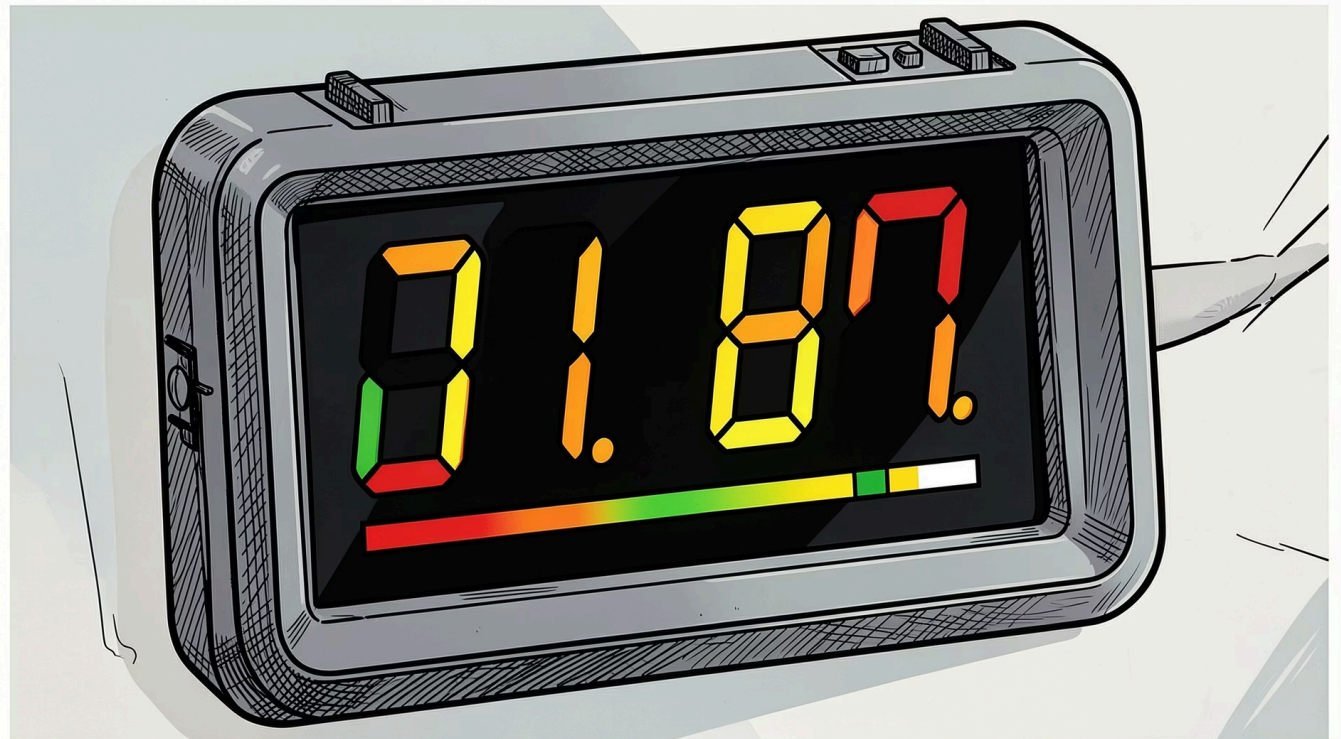
Q1–Q2: Outcomes & How We Track Them

Q1: Expected Measurable Outcomes

- **99% Cost Savings** — Local execution via Ollama drops external API cost per task to \$0
- **85% Token Reduction** — JIT AST rotation & Log Evaporation collapse context sizes
- **Sub-1ms Security** — AST-level gas metering blocks infinite loops in under 1ms

Q2: Tracking Framework

- **Devotion Score** ($D \geq 0.85$) — Evaluates execution speed & token efficiency
- **Goal Drift Index (GDI)** — Real-time semantic tracking; alarm if $GDI > 0.35$
- **STAI Metrics** — Validates code integrity before commits (85% similarity threshold)



Q3: Business Value & Platform Metrics

Value for Developers

A **100% private, local-first** code assistant that auto-heals bugs and unit tests.

Value for Redrob Platform

A **zero-hosting-risk** assessment environment that grades submissions without server exposure.

100%

Local Execution

<1ms

Loop Threat Block

\$0

Execution Cost

99.9%

State Recovery